E-Commerce Customer Segmentation

Project Report

Using K-Means, Agglomerative Clustering, DBSCAN & GMM

UCI Online Retail Dataset | 4,338 Customers | 3 Segments Identified

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1. Executive Summary

This report documents a complete customer segmentation project conducted on a UK-based e-commerce retailer's transactional dataset. The objective was to identify distinct customer groups based on purchasing behaviour, enabling the business to move away from one-size-fits-all marketing towards precision-targeted strategies.

Using the RFM (Recency, Frequency, Monetary) framework and four clustering algorithms — K-Means, Agglomerative Hierarchical Clustering, DBSCAN, and Gaussian Mixture Models — we identified three well-defined customer segments from 4,338 customers. K-Means, validated as the best-performing algorithm on all three evaluation metrics, produced the following segments:

Segment	Customers	Recency	Frequency	Avg. Spend	Description
Champions	1,367 (32%)	30 days	9.6 orders	£5,388	Most valuable — recent, frequent, high-spend VIPs
Loyal Customers	1,984 (46%)	54 days	2.0 orders	£577	Regular buyers — moderate spend, growth opportunity
Occasional Buyers	987 (23%)	255 days	1.4 orders	£406	Lapsed customers — need win-back campaign

> Key finding: 32% of customers (Champions) generate disproportionately high revenue. Retaining and rewarding this group should be a business priority. The 23% in the Occasional Buyers segment represents a significant re-engagement opportunity.

2. The Logic Behind Customer Segmentation

2.1 Why Businesses Use Segmentation

Customer segmentation is the practice of dividing a customer base into groups of individuals that share similar characteristics. Rather than treating every customer identically, businesses use segmentation to ensure that the right message reaches the right person at the right time — maximising the impact of every pound spent on marketing.

The case for segmentation is rooted in a simple observation: customers differ enormously in their behaviour, needs, and value to the business. A customer who orders weekly and spends thousands of pounds per year has fundamentally different needs and sensitivities than one who made a single purchase two years ago. Treating them the same wastes budget and drives disengagement.

2.2 How Segmentation Creates Business Value

Targeting and Personalisation

Once customers are grouped, every communication — email subject lines, ad creative, offer type, timing — can be tailored to that group's behaviour. Personalisation consistently outperforms generic messaging across open rates, click-through rates, and conversion rates.

- Champions receive loyalty rewards and VIP early access
- Loyal Customers receive upsell recommendations aligned to their purchase history
- Occasional Buyers receive win-back campaigns with time-limited incentives

Resource Optimisation

Not all customers are equally worth pursuing. Segmentation allows businesses to allocate budget proportionally to customer lifetime value — spending more to retain high-value customers and less on campaigns unlikely to convert low-engagement groups.

Product and Pricing Strategy

Segment profiles reveal which products are popular with which groups, enabling smarter inventory planning, bundling strategies, and dynamic pricing. High-frequency buyers are ideal targets for subscription or bundle offers; low-frequency, lapsed customers may respond better to a single discounted item.

2.3 Theoretical Foundation

Types of Segmentation

- Demographic segmentation: Age, gender, location, income — useful for consumer goods
- Behavioural segmentation: Purchase history, browsing patterns, product affinity — most relevant for e-commerce
- Psychographic segmentation: Lifestyle, values, attitudes — used in brand strategy
- Transactional segmentation (our approach): Derived entirely from purchase data — objective, data-driven, and highly predictive of future behaviour

Clustering as the Computational Method

Segmentation is operationalised through unsupervised machine learning — specifically, clustering algorithms. Clustering groups data points such that those within a group are more similar to each other than to those in

other groups. The key challenge is defining 'similarity', which is where distance metrics and feature engineering become critical.

The RFM Framework

RFM is the industry-standard framework for behavioural customer segmentation in retail and e-commerce. It characterises each customer along three dimensions derived directly from their transaction history:

- Recency (R): How recently did the customer make a purchase? Recent buyers are more likely to respond to outreach.
- Frequency (F): How many times has the customer purchased? Frequent buyers signal loyalty and engagement.
- Monetary (M): How much has the customer spent in total? This directly reflects customer lifetime value.

> RFM is powerful because it is entirely derived from transactional data — no surveys, no demographics, no assumptions. It measures revealed behaviour, not stated preferences.

3. The Data

3.1 Dataset Overview

The dataset used in this project is the UCI Online Retail Dataset, a widely used benchmark dataset in customer analytics. It contains all transactions occurring between December 2010 and December 2011 for a UK-based online retailer that sells giftware and home accessories.

- Source: UCI Machine Learning Repository
- Total records: 541,909 rows
- Columns: 8
- Data type: Transactional (one row per invoice line item)

The eight columns are as follows:

Column	Type	Description
InvoiceNo	String	Unique invoice identifier. Prefix 'C' denotes a cancellation.
StockCode	String	Product code. Not used in segmentation.
Description	String	Product description. Not used in segmentation.
Quantity	Integer	Units per transaction. Negative values indicate returns.
InvoiceDate	Datetime	Date and time of the transaction. Used to calculate Recency.
UnitPrice	Float	Price per unit in GBP sterling.
CustomerID	Float	Unique customer identifier. 24.93% of rows have no CustomerID (anonymous transactions).
Country	String	Country of the customer. Predominantly United Kingdom.

3.2 Data Preparation

Step 1 — Dropping Missing CustomerIDs

135,080 rows (24.93%) lacked a CustomerID, meaning they represented anonymous transactions that could not be attributed to any individual customer. These rows were removed, as customer-level segmentation requires a customer identifier.

Step 2 — Removing Cancellations

8,905 rows with InvoiceNo beginning with the letter 'C' were removed. Cancellations represent reversed transactions and would distort both frequency and monetary calculations if retained.

Step 3 — Removing Invalid Quantities and Prices

40 rows with Quantity or UnitPrice less than or equal to zero were removed. These represent data entry errors or system-generated records rather than genuine customer purchases.

Step 4 — Deriving TotalPrice

A new column, TotalPrice, was computed as Quantity multiplied by UnitPrice. This forms the basis of the Monetary dimension in the RFM model.

Post-Cleaning Summary

- Remaining records: 397,884
- Unique customers: 4,338
- Date range: December 2010 to December 2011

3.3 Assumptions and Constraints

- The snapshot date used to calculate Recency was set to one day after the last recorded transaction, treating that point as "today".
- All customers are treated equally regardless of country. A UK-specific analysis would yield different segment proportions.
- The dataset covers exactly one year. Seasonality effects (notably the spike in November–December) influence the Recency calculations for customers who purchased in that period.
- DBSCAN identified 114 customers as statistical outliers. These ultra-high-value whale customers were excluded from the main segment profiles but are noted separately.

4. Concepts Applied

4.1 Clustering Algorithms

Four clustering algorithms were evaluated in this project. Each approaches the problem of finding natural groupings differently, with different assumptions about cluster shape, density, and the treatment of outliers.

Algorithm	Cluster Shape	Handles Outliers	Scalability	Result in Project
K-Means	Spherical / convex	No — sensitive to outliers	Excellent	Winner (Sil 0.42)
Agglomerative	Arbitrary shape	Moderate	Moderate	2nd (Sil 0.38)
DBSCAN	Arbitrary shape	Yes — labels noise	Good	Found 114 outliers
GMM	Elliptical / soft	Moderate	Good	3rd (Sil 0.19)

K-Means

K-Means is an iterative partitioning algorithm. It works by randomly initialising K cluster centroids, assigning every data point to the nearest centroid, updating each centroid to the mean of its assigned points, and repeating until convergence. The algorithm minimises inertia — the total sum of squared distances from each point to its centroid.

K-Means assumes clusters are spherical and roughly equal in size, and is sensitive to outliers because the mean is not a robust statistic. After applying a log-transform to correct the extreme skew in the RFM data, K-Means proved to be the strongest performer in this project.

Agglomerative (Hierarchical) Clustering

Agglomerative clustering builds a hierarchy of clusters from the bottom up. It begins by treating every data point as its own cluster, then iteratively merges the two closest clusters according to a linkage criterion. We used Ward linkage, which minimises the total within-cluster variance at each merge step.

The output can be visualised as a dendrogram — a tree diagram showing how clusters are nested within one another. This is particularly useful for choosing the number of clusters visually, as natural breaks appear as long vertical branches in the dendrogram.

DBSCAN (Density-Based Spatial Clustering of Applications with Noise)

DBSCAN defines clusters as dense regions of points separated by sparser regions. It requires two parameters: epsilon (eps), the radius of the neighbourhood around each point, and min_samples, the minimum number of points required to form a dense region. Points that do not belong to any dense region are labelled as noise (cluster -1).

DBSCAN does not require specifying K in advance and naturally identifies outliers. In this project, after tuning eps to 0.3 based on the k-distance elbow plot, DBSCAN identified 2 clusters and flagged 114 customers (2.6%) as noise — these are the ultra-high-value whale customers whose behaviour is statistically anomalous and does not fit the general patterns.

Gaussian Mixture Models (GMM)

GMM is a probabilistic model that assumes data is generated from a mixture of Gaussian (normal) distributions. Unlike K-Means, which makes hard assignments (each point belongs to exactly one cluster), GMM produces

soft assignments — each customer receives a probability of membership in each cluster. A customer might be 70% Champion and 30% Loyal Customer, for example.

In this project, GMM underperformed relative to K-Means (Silhouette of 0.19 vs 0.42). This indicates that the clusters, after log-transformation and robust scaling, became compact and well-separated — a shape that favours K-Means over a probabilistic model designed for overlapping distributions.

4.2 Distance Metrics

All algorithms in this project used Euclidean distance as the measure of similarity between customers in RFM feature space. Euclidean distance between two points A and B in n dimensions is the straight-line distance: the square root of the sum of squared differences across all dimensions.

Euclidean distance is appropriate here because our features — Recency, Frequency, and Monetary — are all continuous numerical variables that have been scaled to comparable ranges. Without scaling, Monetary values in the thousands would dominate the distance calculations, making Recency and Frequency effectively irrelevant.

> For datasets with high dimensionality (many features) or categorical variables, other distance measures such as cosine similarity or Gower distance may be more appropriate. With three well-scaled RFM features, Euclidean distance is the natural choice.

4.3 Dimensionality Reduction — PCA

Principal Component Analysis (PCA) was used for visualisation purposes. Because the RFM feature space has three dimensions and human perception is limited to two or three, PCA projects the data onto a lower-dimensional space that captures the maximum variance.

PCA computes new axes (principal components) that are linear combinations of the original features, ordered by the amount of variance they explain. The first two components typically capture the dominant structure in the data. All PCA scatter plots in this project were produced using two components, with the percentage of variance explained noted on each axis.

It is important to note that PCA was used only for visualisation, not for clustering. Clustering was performed on the full three-dimensional scaled RFM space to preserve all information.

4.4 Evaluation Methods

Metric	What It Measures	Ideal Value	Our Result
Silhouette Score	How well-separated clusters are (−1 to +1)	Closer to +1	0.42 (K-Means)
Davies-Bouldin Index	Avg similarity of each cluster to its most similar cluster	Lower is better	0.81 (K-Means)
Calinski-Harabasz	Ratio of between-cluster to within-cluster dispersion	Higher is better	4,591 (K-Means)

Elbow Method

The elbow method plots K-Means inertia (total within-cluster sum of squares) against the number of clusters K. As K increases, inertia decreases — more clusters means each cluster is smaller and tighter. The 'elbow' is the point where adding more clusters yields diminishing returns. In our results, K=3 clearly marked the elbow, with silhouette score peaking at 0.4218 for K=3 before declining.

BIC / AIC for GMM

For Gaussian Mixture Models, the Bayesian Information Criterion (BIC) and Akaike Information Criterion (AIC) serve a similar purpose to the elbow method. Both penalise model complexity. The minimum BIC/AIC identifies the optimal number of components. In our results, BIC turned negative after K=4 and continued declining, a sign that GMM was overfitting the log-transformed data at higher K values.

K-Distance Plot for DBSCAN

The k-distance plot sorts all data points by their distance to their K-th nearest neighbour. An elbow in this plot indicates the optimal value of eps: below the elbow, points are in dense regions (core points); above it, they are in sparse regions (noise). Our k-distance plot showed an elbow at approximately 0.3, which was used as the eps parameter.

5. Project Steps

Step 1 — Data Collection and Preprocessing

The raw transactional data was loaded from a CSV file using a custom data_loader.py module with configurable delimiter and encoding parameters. The file required latin1 encoding due to special characters in product descriptions.

Preprocessing was handled by preprocessing.py, executing the cleaning steps described in Section 3.2: dropping anonymous transactions, removing cancellations, filtering invalid values, and deriving TotalPrice. After cleaning, 397,884 valid transaction rows representing 4,338 unique customers remained.

Step 2 — Exploratory Data Analysis

EDA was conducted using eda.py and covered four key areas:

- Data quality audit: Null counts and percentages for each column revealed the 24.93% missing CustomerID rate as the primary data quality issue.
- Numeric distributions: Histograms and boxplots of Quantity and UnitPrice revealed extreme right skew — a small number of bulk orders with very high quantities and prices. This flagged the need for outlier-robust preprocessing.
- Revenue over time: Monthly revenue showed a clear seasonal spike in November 2011, consistent with pre-Christmas gifting behaviour.
- Geography: The United Kingdom accounted for the vast majority of revenue. Germany, France, EIRE, and the Netherlands were the leading non-UK markets.

Step 3 — Feature Engineering (RFM)

The transactional data was aggregated to the customer level to produce one row per customer with three features:

- Recency: Days between the customer's last purchase and the snapshot date (December 10, 2011 + 1 day). Lower values indicate more recent activity.
- Frequency: Count of unique InvoiceNo values per customer. This measures how many distinct shopping occasions the customer had.
- Monetary: Sum of TotalPrice per customer. This measures cumulative spend.

The raw RFM distributions were heavily right-skewed: the majority of customers had low frequency and monetary values, while a small number of highly active customers had extreme values (e.g., max Monetary of £280,206). Left untreated, these extremes would dominate the clustering.

Two transformations were applied to correct this:

- Log1p transformation: Applied to Frequency and Monetary. This compresses the right tail without removing the data, making the distributions approximately normal.
- RobustScaler: After log-transformation, RobustScaler was applied to standardise each feature using the median and interquartile range rather than the mean and standard deviation — a more outlier-resistant approach than StandardScaler.

> The impact of these transformations was significant: the initial run without them produced a severely imbalanced segmentation where 93% of customers fell into a single cluster. After transformation, the three segments were distributed 46% / 32% / 23% — a far more actionable result.

Step 4 — Model Selection and Clustering

All four algorithms were fitted on the scaled RFM feature matrix. For K-Means and Agglomerative Clustering, K=3 was selected based on the elbow/silhouette analysis. For GMM, K=3 was selected based on the BIC/AIC plot. For DBSCAN, eps=0.3 was selected from the k-distance plot, with min_samples=10.

Each algorithm produced a cluster label for every customer. DBSCAN additionally produced -1 labels for 114 noise points, which were excluded from the comparative evaluation.

Step 5 — Validation and Refinement

The four label sets were evaluated using three metrics: Silhouette Score, Davies-Bouldin Index, and Calinski-Harabasz Index. K-Means ranked first on all three metrics, confirming it as the chosen model.

An important refinement was made to the segment labelling logic. An initial heuristic that compared each cluster's values to the dataset median produced duplicate labels — two clusters were both called 'Champions'. This was replaced with a rank-based labelling system that computes a composite RFM score per cluster (combining individual dimension ranks) and assigns labels according to overall rank, guaranteeing unique segment names regardless of the number of clusters.

Step 6 — Profiling and Labelling Customer Segments

The final cluster profiles were computed by grouping the RFM table by cluster label and computing the mean of each RFM dimension. The three segments were labelled as follows:

- Champions (Cluster 1): Mean Recency 30 days, mean Frequency 9.6 orders, mean Monetary £5,388. These are the business's most engaged and valuable customers.
- Loyal Customers (Cluster 0): Mean Recency 54 days, mean Frequency 2.0 orders, mean Monetary £577. Moderate across all dimensions — these customers are engaged but not maximising their potential.
- Occasional Buyers (Cluster 2): Mean Recency 255 days, mean Frequency 1.4 orders, mean Monetary £406. These customers have largely disengaged and are at risk of permanent churn.

> DBSCAN's 114 noise points are a distinct fourth group not included in the main profiles. These whale customers — characterised by extraordinarily high frequency and monetary values — are outliers in the statistical sense and warrant individual account management rather than automated campaign targeting.

Step 7 — Translating Results into Actionable Insights

The final step was mapping each segment to concrete business actions — described in full in Section 6. The key principle is that each segment has a different primary goal: retain Champions, grow Loyal Customers, and re-engage Occasional Buyers.

6. How to Use the Results

6.1 Segment-Level Marketing Strategy

Segment	Goal	Tactics	Channel
Champions	Retain & reward	Loyalty rewards, early access, VIP events	Email, personal outreach, exclusive portal
Loyal Customers	Grow spend & frequency	Bundle offers, cross-sell, upsell campaigns	Email, retargeting ads, on-site recommendations
Occasional Buyers	Re-engage & recover	Win-back discount, "We miss you" sequence	Email, SMS, paid re-engagement campaigns

Champions — Retain and Reward

Champions are the business's most profitable customers and the hardest to replace. The strategic priority is retention. Because they are already highly engaged, aggressive discounting is counterproductive — it erodes margin without adding value. Instead, the focus should be on recognition, exclusivity, and deepening the relationship.

- Launch a tiered loyalty programme with meaningful non-discount rewards: early product access, personalised thank-you notes, dedicated customer service.
- Invite Champions to product feedback panels or advisory groups — this both flatters and deepens engagement.
- Identify the products most frequently purchased by Champions and ensure they are always in stock and prominently featured.
- Monitor Champions closely for signs of recency increase — any Champion whose Recency exceeds 90 days should trigger an automated re-engagement flow.

Loyal Customers — Grow Spend and Frequency

Loyal Customers are the largest segment (46%) and represent the greatest growth opportunity. They are already engaged enough to purchase repeatedly, but their average order value and frequency are significantly below Champions. The goal is to increase both.

- Use cross-sell and upsell campaigns based on purchase history — 'Customers who bought X also bought Y'.
- Offer bundle deals that increase the average order value per transaction.
- Introduce a frequency incentive: 'Buy 3 times in 60 days and receive a reward' to accelerate their progression toward Champion status.
- A/B test personalised email subject lines that reference their actual purchase history to increase open rates.

Occasional Buyers — Re-engage Before They Churn

With a mean recency of 255 days, Occasional Buyers are close to fully lapsed. The window for re-engagement is narrow and the cost-per-recovery is relatively high, so campaigns targeting this segment need strong incentives and clear urgency.

- Deploy a multi-touch win-back sequence: (1) 'We miss you' email with a time-limited 15% discount, (2) SMS follow-up after 7 days if not opened, (3) Final 'last chance' email before suppression.
- Suppress persistent non-responders after 3 touchpoints to protect sender reputation and reduce wasted spend.
- For those who do respond, enrol them in the Loyal Customer journey to rebuild frequency.

6.2 Product Recommendations

The rfm_with_clusters.csv output links each CustomerID directly to their segment. This file can be joined to the original transaction data to produce segment-level product affinity analysis:

- Identify the top 10 products by revenue within each segment — these become the hero products for each segment's campaigns.
- Identify products purchased exclusively by Champions — these signal aspirational or premium items that could be used to motivate Loyal Customers to upgrade.
- Identify products with high Occasional Buyer affinity — these are useful as 'anchor products' in win-back campaigns, since these customers have demonstrated a preference for them in the past.

6.3 Customer Retention and Loyalty Programmes

A key insight from the segmentation is that customer value is highly concentrated. The 1,367 Champions (32% of customers) likely account for a substantially higher share of total revenue, while the 987 Occasional Buyers contribute a much smaller proportion despite being a meaningful share of the customer base.

Retention programmes should be tiered accordingly:

- Premium tier (Champions): High-touch, personalised, proactive. Individual outreach from account managers for highest-value customers within this group.
- Standard tier (Loyal Customers): Automated but personalised. Behavioural triggers (lapsed days, basket abandonment, category browsing) drive communications.
- Reactivation tier (Occasional Buyers): Time-boxed win-back campaign with clear suppression rules. Measure recovery rate and revenue-per-email to assess ROI.

6.4 Strategic Decision-Making

Pricing

Champions demonstrate low price sensitivity — they purchase frequently at full price. Loyalty discounts for this group should be used sparingly and framed as rewards, not necessity. Occasional Buyers, by contrast, are more likely to respond to promotional pricing as a re-engagement mechanism.

Promotions and Campaign Planning

Segment the promotional calendar by customer group. Blanket promotions sent to all customers simultaneously devalue the offer for Champions while being insufficient to re-engage Occasional Buyers. Staggered or exclusive promotions by segment produce better results at lower margin cost.

Inventory and Supply Chain

Champion-preferred products should be prioritised in stock planning. A stockout on a product purchased regularly by Champions causes disproportionate brand damage relative to the same stockout for a lower-engagement segment.

Resource Allocation

CRM and marketing investment should be proportional to segment value and opportunity. The typical allocation principle: 40–50% of CRM budget on Champion retention, 35–45% on Loyal Customer growth, and 10–20% on Occasional Buyer re-engagement, adjusting based on measured recovery rates.

> The segment labels are not permanent. Customers should be re-segmented quarterly or after any major promotional event. A customer who responds to the win-back campaign should graduate to Loyal Customer status in the next cycle, not continue receiving Occasional Buyer messaging.

7. Conclusions and Recommended Next Steps

7.1 What We Achieved

This project progressed from a raw transactional dataset to a fully validated, business-interpretable customer segmentation in a structured, reproducible pipeline. Key achievements:

- Cleaned and processed 541,909 transaction rows to produce a 4,338-customer RFM table.
- Applied log-transformation and robust scaling to resolve a severe cluster imbalance (from 93% in one cluster to a balanced 46/32/23 split).
- Evaluated four clustering algorithms against three validation metrics — K-Means was confirmed as the best performer.
- Produced three distinct, uniquely labelled segments with clear business interpretation.
- Identified 114 statistical outlier customers (DBSCAN noise) representing an ultra-high-value whale segment requiring individual attention.

7.2 Limitations

- One year of data: The dataset covers a single calendar year, so long-term behavioural patterns and true customer lifetime value cannot be assessed. A multi-year dataset would enable churn prediction and cohort analysis.
- No demographic data: The segmentation is entirely behavioural. Combining RFM with demographic features (age, location, channel preference) would produce richer segments.
- Static snapshot: Segments reflect a point-in-time analysis. Customer behaviour shifts over time and segmentation should be re-run periodically.

7.3 Recommended Next Steps

1. Re-segment quarterly using the same pipeline — update rfm_with_clusters.csv and track segment migration rates (how many Occasional Buyers graduate to Loyal Customers each quarter).
2. Enrich segments with product-level data — join cluster labels to the transaction data to compute product affinity by segment.
3. Build a churn prediction model — use the Occasional Buyer segment as the positive class in a binary classifier to identify customers at risk before they fully lapse.
4. A/B test the segment-specific campaigns — measure whether personalised messaging to each segment outperforms the current generic approach on revenue, open rate, and repeat purchase rate.
5. Extend to geographic sub-segmentation — the top non-UK markets (Germany, France, Netherlands) may warrant their own segment profiles given different purchasing seasonality and product preferences.

Appendix — Technical Reference

A. Algorithm Comparison Results

Algorithm	N Clusters	Silhouette ↑	Davies-Bouldin ↓	Calinski-H ↑	Noise %
K-Means	3	**0.4218**	**0.8069**	**4,591.3**	0.0%
Agglomerative	3	0.3805	0.8580	3,875.8	0.0%
DBSCAN	2	0.3210	1.1822	2,282.6	2.6%
GMM	3	0.1935	1.5679	2,116.9	0.0%

B. Project File Structure

The project is organised as a modular Python package under Cust_Seg/:

File	Purpose
main.py	Entry point. Runs the full pipeline end-to-end.
src/data_loader.py	Loads CSV with configurable encoding and delimiter.
src/preprocessing.py	Cleans data, builds RFM, applies log-transform and scaling.
src/eda.py	Exploratory analysis: null audit, distributions, revenue and country charts.
src/clustering.py	All four algorithms, diagnostic plots, comparison, and rank-based segment labelling.
src/visualization.py	PCA scatter, snake plot, heatmap, cluster size bar chart.
outputs/	All generated plots (.png), cluster_profile.csv, rfm_with_clusters.csv, algorithm_comparison.csv.

C. Configuration Parameters

The following parameters in the CONFIG block of main.py can be tuned after reviewing diagnostic plots:

Parameter	Current Value	How to Choose
K_MEANS_N	3	Review elbow/silhouette plot — choose K at the elbow with highest silhouette.
AGGLO_N	3	Review dendrogram — choose K where the longest vertical branches appear.
GMM_N	3	Review BIC/AIC plot — choose K at the minimum BIC.
DBSCAN_EPS	0.3	Review k-distance plot — choose the distance value at the elbow.

Parameter	Current Value	How to Choose
DBSCAN_MIN_SAMP	10	Rule of thumb: 2 x number of features. Increase for larger datasets.